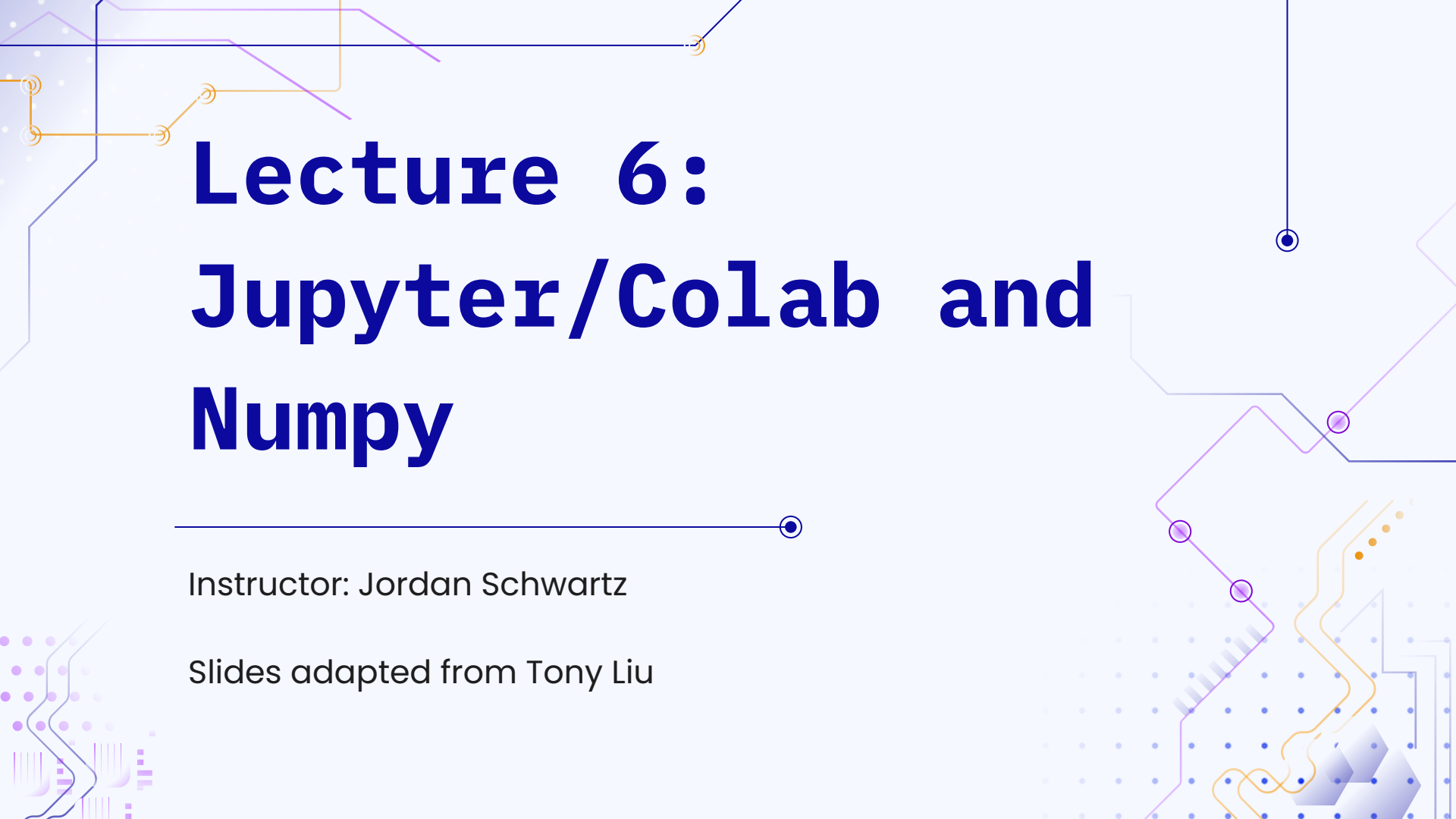




Lecture 6: Jupyter/Colab and Numpy

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Slides adapted from Tony Liu

Logistics

- HW2 due this morning
 - How did that go?
- HW3 released sometime today

Week 6

- Jupyter/Colab
- Numpy
- Brief intro to matplotlib

What is module 3, Data Science, about?

- **Lecture 6 - Jupyter/colab and Numpy**
 - What is data science
 - What are notebooks
 - Jupyter
 - How to open a jupyter notebook vs a colab notebook
 - Intro to numpy
 - Broadcasting and vectorization
 - If time- Intro to Matplotlib
 - Just a high level overview, go in depth next lecture

What is module 3, Data Science, about?

- Lecture 7 - Pandas, Matplotlib, EDA
 - What is pandas
 - What can pandas do
 - What is a dataframe
 - How to create a dataframe
 - EDA example
 - Seaborn plotting
- Lecture 8 - Machine Learning
 - What is machine learning?
 - Different types of machine learning
 - Defining a model
 - Disclaimer about math
 - Break it into:
 - supervised learning
 - Classification
 - Regression
 - The full machine learning process
- Lecture 9 - Deep Learning
 - NN inspiration and perceptrons
 - Activation functions
 - What are NNs
 - Hidden layers and training
 - Loss functions and weights
 - Gradient descent
 - Backpropagation
 - PyTorch or Keras example



What is Data Science about?

Data Science

- In short, data science is the practice of answering questions about the world using collected data
- Data scientists use insights about data to build models to automate certain tasks
 - Machine learning = tuning models
- These models are driven by *statistical patterns* found in the data

ML isn't magic, it's math

- Machine learning is typically treated like magic, but its foundations lie in statistics
- At the end of the day, it's an experimental science
 - Pick some parameters, see how "good" they are, repeat

What we are building up to...

Typically, the flow for training a model is something like:

1. Data Collection
2. Data Preprocessing
3. Model Training
4. Model Evaluation
5. Model Deployment

where steps 3-4 could be run many times with different classes of models.

For now...

- For less complex models, the arguably best choice of tools is to use for steps 1-4 are the `numpy` and `pandas` Python modules
- We won't be focusing on optimizing steps 3-4, but we will go over a few types of models you can use
 - If you want to learn more about this, take a machine learning course!
- Step 5 is a problem for a data engineer, someone who productionizes models and data pipelines
 - You will do something similar to this for HW3/4!

Notebooks: Jupyter and Colab

- Another way of running Python code
 - but provide some properties that are highly useful for data science
- Code can be logically grouped,
- Outputs can be saved without rerunning code
- Plotting data is very quick and easy
- If you haven't already, install Jupyter with `pip install jupyterlab`

Jupyter and Colab

- Jupyter (<https://jupyter.org/>)
- Colab (<https://colab.research.google.com/>)
- An open-source collection of tools for interactive scientific computing and data science
- Most prominent tool:
 - allows for the development, documentation, execution, and sharing of code
 - two components:
 - .ipynb notebook files
 - browser app for rendering notebooks the Jupyter notebook

Anatomy of a notebook

- Notebook cells of various types:
 - Markdown → rendered text
 - Code (depending on kernel) → in this class they are IPython cells
- Can install Jupyter with
 - `pip install jupyterlab`

Running Jupyter notebooks

- Can install jupyter on your computer, run a local server using jupyter notebook command
- For the purposes of this course, I don't really care what platform you use to host a notebook
 - The two most common I've seen are running a local server and Google Colab
- A common workflow misunderstanding: **cells do not run from top to bottom!**
 - instead, order of execution is on a cell by cell basis

Check in - cell order

What gets printed for...

S7

[5]
✓ Os

1 x = 3

[2]
✓ Os

1 y = 5

[3]
✓ Os

1 x = 10

[4]
✓ Os

1 print(x + y)

S8

[6]
✓ Os

1 x = 3

[2]
✓ Os

1 y = 5

[3]
✓ Os

1 x = 10

[7]
✓ Os

1 print(x + y)



Numpy

What is Numpy?

- Numpy is Python's scientific computing package
- Think complex mathematical functions, matrixes, probability, etc
- We mentioned that Python is relatively slow, so Numpy leverages underlying C code to ensure fast computation
 - However, everything is abstracted away and the code you usually have to write is very minimal!

Numpy

- The main data structure in Numpy is an N-dimensional array: `ndarray`
- In scientific computing, you almost always deal with *vectorized* computation
- As a result, these types of operations are first-class citizens in Numpy

```
# regular python  
c = []  
for i in range(len(a)) :  
    c.append(a[i]*b[i])
```

```
import numpy as np  
# using Numpy ndarrays!  
c = a * b  
# and it's super fast!
```

ndarray

- `np.array(seq)`: copy elements of `seq` to array
- `np.zeros(shape)`, `np.ones(shape)`: arrays of zeros/ones with the given fixed shape
- `np.zeros_like(a)`, `np.ones_like(a)`: arrays of zeros/ones that matches the shape of array `a`
- the different dimensions of an `ndarray` are called axes
- for 2D shapes, first axis is the row and the second is the column

Switch to Colab

<https://colab.research.google.com/drive/1ZU84otfV3uN3KaEr9SyblAnNRMf4-m0G#scrollTo=HLubWol7IROe>

Thanks !

Do you have any questions?

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